Beyond the Capital: Diverging Rents and Sale Prices During Conflict in Iran

Mahyar Rezazadeh*

Abstract

Purpose – Conflict-related population movements may change housing demand in receiving cities, but rents and sale prices may respond differently. This study examines this contrast in Iran during 2025–2026.

Design/methodology/approach – The study estimates hedonic difference-in-differences models using online listings from 2024 to October 2026. A stacked design compares directly exposed neighbourhoods in Tehran (the capital) with nearby areas without recorded exposure. It also compares four Caspian coastal cities (potential relocation destinations) with Tehran, neighbouring Alborz Province and inland cities, using wild-bootstrap and permutation inference.

Findings – Caspian asking rents rose by 27 per cent relative to Tehran and Alborz and 31 per cent relative to inland cities. The 2025 premium subsequently eased; adjusting for differential trends roughly halves the estimated overall increase. Caspian sale asking prices showed no comparable increase relative to inland cities but were approximately 12 per cent higher relative to Tehran and Alborz. Directly exposed Tehran neighbourhoods showed no detectable discount, although moderate declines cannot be excluded. The rental–sale price divergence is consistent with temporary demand but also slower sale-price adjustment or supply responses.

Research limitations/implications – The data record asking rather than transaction prices. Destination-market estimates cover four cities; early weekly samples are small.

Practical implications – Housing-market assessments may consider rental markets in receiving cities alongside directly exposed areas.

Originality – This study jointly examines housing-market adjustments in areas directly exposed to conflict and potential relocation destinations. By comparing rental and sale prices across neighbourhoods and cities, it provides new evidence on how housing-market responses differ across locations and market segments during periods of population movement.

Keywords: Housing markets; asking rents; sale prices; population movements; armed conflict; Iran.

Article classification: Research paper.

JEL classification R21, R31, D74, O18

*PhD Candidate, Department of Economics, Simon Fraser University. Email: smr20@sfu.ca

1 Introduction

When households relocate during armed conflict, housing markets may adjust in more than one location. Housing demand may decline in areas directly exposed to military activity while increasing in cities receiving relocating households. Yet these changes need not appear equally in rents and sale prices. Rents reflect current demand for housing services, whereas sale prices also depend on expectations about future demand. A temporary increase in housing demand may therefore raise rents without producing a comparable increase in property values. Where these adjustments occur, and how rental and ownership markets respond, remains an empirical question with implications for understanding housing-market dynamics during periods of population movement.

Iran's conflicts in 2025 and 2026 provide a setting in which to examine these differences. Tehran, the country's capital and largest urban centre, experienced geographically concentrated military activity, while cities along the Caspian coast in northern Iran became potential destinations for households relocating from the capital. These developments suggest two possible housing-market responses. Within Tehran, direct exposure may have changed the relative attractiveness of affected neighbourhoods. In northern receiving markets (along with Caspian Sea), additional demand for accommodation may have increased rents, particularly if relocating households initially preferred renting to purchasing property. Whether these responses occurred, and whether they differed between rental and sale markets, is the focus of this paper.

This paper examines both dimensions using property listings collected before and after the two conflict periods. Within Tehran, the analysis compares neighbourhoods with recorded direct exposure (struck neighborhoods) to nearby neighbourhoods without such exposure. Across cities, it examines rents and sale prices in four cities in northern Iran's Caspian region (Gilan, Mazandaran and Golestan), where households leaving Tehran may have relocated. Changes in these cities are compared with those in Tehran (the capital), neighbouring Alborz Province and inland cities without recorded direct exposure to military activity. These comparisons make it possible to examine the geographic distribution of housing-market adjustments and whether they are more apparent in locations directly exposed to military activity or in potential relocation destinations.

The analysis draws on Divar, a widely used Iranian online classifieds platform. An openly available dataset contains approximately one million listings posted between May and December 2024, before either conflict. These records are combined with listings collected through a new crawl and two tracking rounds in October 2026, covering 43 cities. The study also compiles geocoded records of military activity from publicly available monitoring sources and match them to neighbourhood boundaries. The resulting repeated cross-sections make it possible to compare asking rents and sale prices across locations and periods while accounting for observed property characteristics.

The study employs two complementary hedonic difference-in-differences designs. For the within-Tehran analysis, each directly exposed neighbourhood is matched with its four nearest neighbourhoods without recorded exposure. These locations are separated by a median distance of 1.3 kilometres. The matched groups are pooled in a stacked specification with stack-specific neighbourhood and period fixed effects, comparing changes within narrowly defined geographic

areas. For the cross-city analysis, the study estimates changes in Caspian housing markets relative to the capital region and inland comparison cities. The sensitivity of the results to alternative comparison groups, weighting schemes and differential pre-existing trends is also examined. Given the small number of Caspian cities, inference incorporates city-level permutation tests and wild-bootstrap procedures.

The clearest finding concerns rental markets in the north. Between 2024 and October 2026, asking rents in the sampled Caspian cities increased by approximately 27 per cent relative to Tehran and Alborz and 31 per cent relative to seven unstruck cities outside the three provinces (inland comparison group). This difference was already apparent in the limited listings available after the June 2025 conflict, when the estimated rental premium reached approximately 9 per cent before subsequently easing. Although these early estimates rely on relatively few observations, they provide suggestive evidence of adjustment following the first conflict period. The positive rental differential also appears across alternative comparison groups and weighting schemes. Allowing for differential trends estimated from the 2024 data reduces the overall premium to approximately 14 per cent. Thus, the estimated magnitude is sensitive to the trend adjustment, but the positive differential is not confined to the baseline specification. The premium stays positive and about 2.7 and 2.0 standard errors from zero even after pre-trend adjustment.

Sale asking prices present a different picture. Relative to inland cities, Caspian sale prices show no comparable increase, although they were approximately 12 per cent higher relative to Tehran and Alborz. Against the inland comparison group, the divergence between rents and sale prices implies a decline in the relative asking-price-to-rent ratio. One possible explanation follows from standard housing asset-pricing models, in which property values reflect the discounted stream of future housing services (Poterba, 1984; Himmelberg *et al.*, 2005). If additional rental demand is expected to be temporary, current rents may increase without a comparable change in sale prices. The easing of the rental premium following the first conflict period is consistent with this interpretation. It is not, however, the only explanation. Slower adjustment of sale listings, changes in housing supply, differences in market uncertainty, or shifts in the composition of advertised properties could produce similar patterns.

Within Tehran, the estimated response is more limited and less precise. Neighbourhoods with recorded direct exposure (struck neighborhoods) show no statistically detectable discount in asking prices or rents relative to their nearest comparison neighbourhoods. The estimated changes are 1.1 per cent for sale asking prices and 2.2 per cent for rents. However, A placebo analysis using the 2024 observations indicates that rents in subsequently exposed neighbourhoods were already increasing somewhat faster than in nearby comparison areas. Extrapolating these differential trends produces moderately negative point estimates, although they remain imprecise. The evidence therefore does not establish a clear local price discount.

Taken together, these findings suggest that housing-market adjustments during the conflicts were more apparent in the rental markets of potential relocation destinations than in the relative asking prices of directly exposed Tehran neighbourhoods. This distinction matters because the geographic distribution of changes in housing demand need not coincide with the locations of direct exposure. It also highlights the importance of distinguishing current housing-service

demand from the valuation of residential property. A rental premium without a comparable sale-price response provides information about market adjustment, even when the persistence of the underlying demand change cannot be established.

Several limitations shape the interpretation of these findings. The data record advertised prices rather than completed transactions or rents actually paid. Changes in listing composition and differences in the frequency with which rental and sale advertisements are updated may influence the estimates. The destination-market analysis also covers only four Caspian cities, while the archived observations from 2025 provide limited information on short-run dynamics. Moreover, the data do not directly reveal household relocation or the expectations of buyers and landlords. The estimated Caspian rental premium is therefore consistent with relocation-related demand, but alternative city-specific changes cannot be fully excluded. These considerations are particularly important when interpreting the divergence between rents and sale prices.

This study relates to three strands of research. The first examines how local exposure to conflict and changes in perceived risk are reflected in housing values. Besley and Mueller (2012) document the relationship between violence and property prices in Northern Ireland, while Elster *et al.* (2017) find persistent housing-price and rental adjustments in Israeli localities exposed to rocket attacks. Related evidence from Mexico City shows that the relationship between local exposure and housing values depends partly on the frequency and visibility of events Atuesta and Carrasco (2024). This paper contributes to this literature by examining price differences between nearby neighbourhoods within a metropolitan area where recorded exposure was geographically concentrated.

The second strand examines housing-market responses to migration and population inflows. Saiz (2007) documents increases in housing rents and values associated with immigration to US cities, while Tumen (2016) finds heterogeneous rental responses to refugee inflows in Turkey. More directly related to internal relocation, Depetris-Chauvin and Santos (2018) shows that displacement inflows in Colombia affected rental prices differently across housing-market segments. Gluszak and Trojanek (2025) examines the arrival of Ukrainian refugees in Polish cities and finds a stronger response in rents than in sale prices. This study builds on this evidence by examining potential internal relocation alongside local exposure within the same national housing market. This joint perspective helps distinguish the geographic distribution of housing-market changes from differences between rental and ownership markets.

Finally, the paper contributes to the broader empirical literature on spatially differentiated housing-market adjustments. Related empirical research examines how housing prices and rents respond to public-health disruptions, local violence, and changes in housing policy (Qin *et al.*, 2023; Atuesta and Carrasco, 2024; Pinto Hernández *et al.*, 2026). The paper contributes to this literature by examining housing-market adjustments both in areas directly exposed to conflict and in potential relocation destinations, while distinguishing between rental and sale markets. A closely related study by Gholipour *et al.* (2026) documents housing-market functioning and emergency housing governance in Tehran during the same two conflict periods. This paper complements that descriptive evidence with neighbourhood-level and cross-city comparisons, including housing markets outside the capital. The central contribution is to bring these geographic and market dimensions together, showing that rental and sale markets may register

different aspects of conflict-related changes in housing demand.

The remainder of the paper is organized as follows. Section 2 develops the conceptual framework and reviews the related literature. Section 3 describes the setting and data. Section 4 presents the empirical strategy. Section 5 reports the main findings and robustness checks. Section 6 discusses possible mechanisms and limitations, and Section 7 concludes.

2 Related literature and conceptual framework

2.1 How violence enters housing prices

In a spatial equilibrium, the price of a location capitalises the expected flow of its amenities and disamenities (Rosen, 1974; Roback, 1982). Violence is a disamenity whose capitalisation depends on what it reveals about the future. Persistent violence lowers prices. The end of the Troubles raised house prices most in the Northern Ireland regions that had suffered most (Besley and Mueller, 2012); rocket fire from Gaza lowered prices in Israeli towns within range, and the Iron Dome system reversed part of the loss (Elster *et al.*, 2017; 2019). A single dramatic event can also shift prices when it changes beliefs about a place, as the murder of Theo van Gogh did in Amsterdam neighbourhoods with large Muslim populations (Gautier *et al.*, 2009), while the September 2001 attacks raised office vacancy rates around Chicago's landmark buildings (Abadie and Dermisi, 2008). The broader economic costs of conflict are well documented (Abadie and Gardeazabal, 2003), and the long-run evidence from bombing campaigns points to strong recovery of population and activity in bombed cities (Davis and Weinstein, 2002; Brakman *et al.*, 2004; Miguel and Roland, 2011), though not always of agglomeration economies (Dericks and Koster, 2021).

Studies of environmental and health risks reach a similar conclusion through different data: markets discount a location when the risk is salient, local and expected to recur, as with cancer clusters (Davis, 2004), sex-offender residence (Linden and Rockoff, 2008), hurricanes (Hallstrom and Smith, 2005) or exposure to flooding (Ortega and Taşpınar, 2018; Bernstein *et al.*, 2019). Physical damage, by contrast, need not leave a lasting mark: economic activity in flooded cities recovers quickly, even where floods recur (Kocornik-Mina *et al.*, 2020).

These results suggest a simple test. Many strikes on Tehran's residential neighbourhoods in June 2025 were assassinations carried out with precision munitions; in 2026 they became far more numerous but remained, close to the residential north Tehran. If buyers inferred that the danger was attached to particular occupants, the struck neighbourhoods would carry little lasting discount relative to their neighbours. If they inferred that the neighbourhood itself had become a target, prices and rents would fall there relative to nearby streets.

2.2 Displacement as a demand shock

Population movements can change housing demand in receiving cities, particularly when the existing housing stock cannot adjust quickly to additional demand. An increase in the number of households seeking accommodation may therefore raise rents, although the magnitude depends on available housing, supply responses and the relocation decisions of existing residents. Evidence from immigration studies illustrates these differences. Saiz (2003) documents rental

increases in Miami following the Mariel boatlift, while Saiz (2007) finds a positive relationship between immigration and housing costs across US cities. However, migration does not necessarily increase prices in every local market. Evidence from the United Kingdom and Italy shows that the movement of existing residents can offset some of the additional demand and alter the distribution of housing-price changes across neighbourhoods (Sá, 2015; Accetturo *et al.*, 2014).

Studies of refugee and internal-displacement inflows offer further evidence of heterogeneous responses. In Turkey, Tumen (2016) finds higher rents for better-quality housing following the arrival of Syrian refugees, with no significant change for lower-quality units. In Colombia, Depetris-Chauvin and Santos (2018) documents rising rents in lower-income housing markets but declining rents in higher-income segments. Evidence from Poland is particularly relevant to the distinction between rental and ownership markets. Following the arrival of Ukrainian refugees, rents increased more strongly than sale prices in receiving cities (Trojanek and Gluszak, 2022; Gluszak and Trojanek, 2025). Related research on New Zealand also links the return of residents during the COVID-19 pandemic to changes in housing prices (Ngo *et al.*, 2025). Together, these studies suggest that population inflows can affect housing markets differently depending on local conditions and the type of housing considered.

The distinction between rents and sale prices is central to understanding these responses. Rents reflect the current demand for housing services, whereas property values depend on the expected stream of future housing services and the cost of holding residential assets (Poterba, 1984; Himmelberg *et al.*, 2005). Holding other factors constant, an increase in demand expected to be temporary may raise current rents without producing a comparable increase in sale prices. A more persistent increase in demand would be more likely to affect both. A declining price-to-rent ratio may therefore be consistent with expectations of temporary demand, but it is not sufficient to establish them. Slower adjustment of sale prices, changes in housing supply and shifts in required returns can produce similar patterns. Moreover, housing-market responses in origin and destination cities need not be symmetric. Because housing is durable, declining demand may reduce property values without generating a comparable reduction in the existing housing stock (Glaeser and Gyourko, 2005). These considerations motivate the separate examination of rental and sale markets in potential relocation destinations and of housing values in directly exposed neighbourhoods.

3 Setting and data

3.1 Conflict periods and housing-market observations

Iran's housing markets experienced two periods of potential demand reallocation associated with the conflicts of June 2025 and February–April 2026. During these periods, households relocating from Tehran and other major cities may have increased demand for accommodation in receiving areas, particularly in northern Iran. The scale of these movements was substantial. According to preliminary estimates from the United Nations High Commissioner for Refugees (UNHCR), between 600,000 and one million households, representing up to 3.2 million people, were temporarily displaced within Iran by March 2026. Many reportedly moved from major urban centres towards northern and rural areas. Although these figures do not identify the

destinations of individual households, they provide context for examining changes in housing demand across locations.

Both conflict periods also coincided with substantial restrictions on internet connectivity, affecting the availability of online housing listings. Internet access was disrupted during June 2025 and was largely restored on 25 June. A further interruption beginning on 28 February 2026 was followed by a partial restoration on 26 May. These interruptions limit the ability to observe housing-market developments continuously. The analysis therefore combines listings from 2024, before either conflict, with observations collected in October 2026. Archived listings from the weeks following the restoration of internet access in June 2025 provide additional, although limited, information about short-run rental-market adjustments.

Inflation is another important consideration when comparing housing prices across periods. Changes in the general price level can affect nominal asking prices and rents across markets simultaneously. The analysis therefore focuses on relative changes across locations, using period fixed effects to account for common time-varying factors. Iranian rental contracts also commonly combine a refundable deposit with a monthly payment. To compare rental listings on a consistent basis, a monthly rent-equivalent is calculated as the advertised monthly rent plus three per cent of the refundable deposit.

3.2 Listings

The study combines three sources of property listings covering the periods before, between and after the two conflicts. Table I summarises the resulting samples.

Before the conflicts. The baseline data come from an openly released Divar dataset containing approximately one million anonymised real-estate advertisements (Divar, 2025). The analysis uses the observations from May to December 2024, before either conflict period. The dataset records asking sale prices or rental deposits and monthly rents, together with property characteristics such as floor area, construction year, number of rooms, building floor, and the availability of elevators, parking and storage. Listings also include neighbourhood identifiers and approximate geographic coordinates. These characteristics make it possible to compare housing prices while accounting for observable differences between properties.

After the conflicts. The author collected active Divar listings on 10 October 2026 in 43 cities, selected because each had at least 800 sale listings in the 2024 data. Two additional tracking rounds on 12 and 14 October identified listings not captured in the initial collection. Unlike the 2024 data, which represent newly posted advertisements over several months, the October 2026 observations provide snapshots of the active listing inventory. The latter also contains a relatively high share of estate-agency advertisements, accounting for approximately four-fifths of the collected listings. These differences in sampling may affect the composition of advertised properties across periods. The analysis therefore focuses on changes in asking prices conditional on observed dwelling characteristics rather than interpreting changes in listing counts as changes in housing supply.

Between the conflicts. To examine the timing of housing-market adjustments, these datasets are supplemented with property pages preserved by the Internet Archive (Internet Archive, 2026). Of 12,956 archived housing pages, 4,966 contained identifiable listings. Coverage is

uneven across time: many pages archived after August 2025 contain incomplete information, while very few listings are available during the 2026 internet restrictions. Consequently, these observations are used primarily to examine rental-market developments during the five weeks following the restoration of internet access in June 2025. The archived sample provides an additional indication of timing but is not treated as a representative continuous panel.

A common cleaning procedure is applied across the three sources. The analysis is restricted to apartments with floor areas between 25 and 400 m^2. Within each city, tenure category and data source, observations are trimmed if they fall below the first percentile or above the ninety-ninth percentile of sale price or monthly rent-equivalent per square metre, as applicable. Building age is calculated from the reported construction year. These restrictions provide a more comparable sample across locations and periods, although differences in unobserved property characteristics may remain.

3.3 Geographic exposure and comparison groups

Neighbourhood-level measures of direct military exposure are constructed using publicly available incident reports. The first source consists of daily reports for the two conflict periods (Human Rights Activists News Agency, 2025; 2026). The author identifies geographic references in these reports and manually matches the reported locations to the transliterated names of Divar's Tehran neighbourhoods. These records are supplemented with a public incident tracker compiled from military statements and news reports (Jewish Institute for National Security of America, 2025). This approach provides a consistent geographic classification based on reported incidents, rather than a complete record of all locations that may have experienced military activity.

Among Tehran's 331 neighbourhoods with housing listings, 42 are identified in the June 2025 records and 9 in the 2026 records, with 50 identified in at least one conflict period. Figure 1 displays their geographic distribution. Recorded exposure is concentrated in parts of northern and eastern Tehran rather than being distributed evenly across the city. Because these areas may differ systematically from other neighbourhoods in housing characteristics and prices, the within-city analysis compares each directly exposed neighbourhood with nearby neighbourhoods without recorded exposure.

The cross-city analysis distinguishes three geographic groups. The first consists of cities in Iran's northern Caspian region, a potential destination for households relocating from Tehran. Although the broader region includes Gilan, Mazandaran and Golestan provinces, the October 2026 sample contains sufficient listings in 4 cities: Rasht, Bandar Anzali and Lahijan in Gilan Province, and Gorgan in Golestan Province (Appendix Figure A1). The second group represents the capital region and includes Tehran and 10 cities in neighbouring Alborz Province and Tehran's surrounding satellite towns. The third group consists of 7 cities located outside the capital region and the northern Caspian group, with no direct military activity recorded in either of the two geographic sources.

The northern cities are treated as potential relocation destinations rather than locations necessarily free of direct exposure. Three of the four sampled cities had one reported incident during the conflict periods. This distinction is relevant because changes in their housing markets

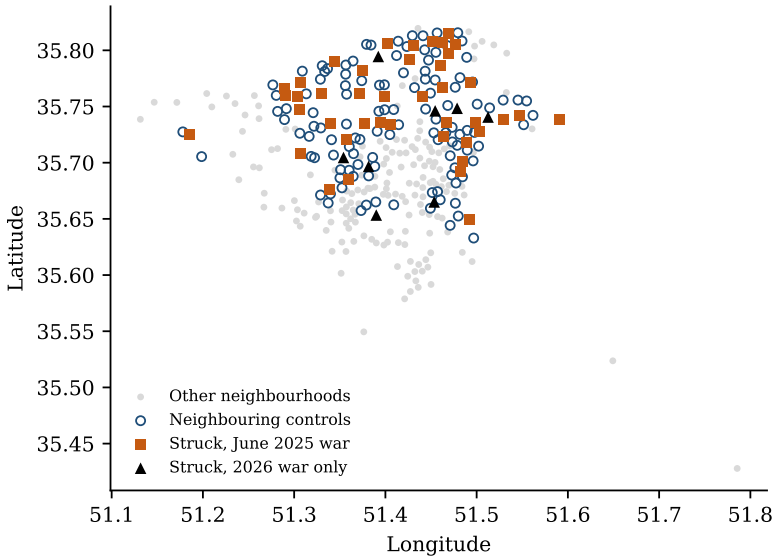


Figure 1: Neighbourhoods with recorded direct exposure and nearby comparison areas, Tehran
Notes: Points represent neighbourhood centroids, calculated using the median coordinates of 2024 property listings. For each neighbourhood with recorded direct exposure, the comparison group includes up to four nearest neighbourhoods without recorded exposure, located within 2.5 km.

may reflect both relocation-related demand and other local conditions. The empirical analysis therefore evaluates the northern rental premium against alternative comparison groups and examines the sensitivity of the estimates to these geographic classifications.

Finally, the study constructs a separate measure of proximity to security-related facilities in Tehran. OpenStreetMap data (OpenStreetMap contributors, 2026) are used to identify police stations and Basij facilities, categories of locations appearing in the 2026 incident reports (Human Rights Activists News Agency, 2026). A neighbourhood is classified as proximate to such a facility if its centroid lies within 500 metres of at least one identified site. Approximately 43 per cent of Tehran's neighbourhoods satisfy this criterion. This measure captures geographic proximity to facilities rather than confirmed direct military exposure, making it possible to examine whether housing-market adjustments differ in areas with these characteristics (Table I).

4 Empirical strategy

4.1 Displacement

For listing i in neighbourhood n of city c in period t, the displacement regression is

$$\ln p_{inct} = \beta\,(\text{Caspian}_c \times \text{Post}_t) + X_i'\gamma + \alpha_n + \lambda_t + \varepsilon_{inct}, \tag{1}$$

where p is the asking price or rent-equivalent per square metre, X_i holds the hedonic controls (log floor area, age and its square, floor, number of rooms, elevator, parking, storage and missing-value indicators), α_n are neighbourhood fixed effects, λ_t are the eight 2024 months and October 2026, and Post marks October 2026. The comparison group is either the capital region or the inland cities without a recorded strike. The identifying assumption is that, absent the wars, Caspian rents and prices would have moved in parallel with those of the comparison cities. The

2024 months allow a test of parallel pre-war movements, and a version of equation (1) with a Caspian-specific linear trend, fitted on 2024 and extrapolated, shows how much the estimate depends on that assumption.

With four treated cities, conventional cluster-robust inference is unreliable (Cameron *et al.*, 2008; MacKinnon and Webb, 2018; Ferman and Pinto, 2019). Two alternatives are reported. The first is the wild cluster restricted bootstrap with Webb weights (Roodman *et al.*, 2019; Webb, 2023). The second is a two-step estimator that treats the city as the unit of observation: listing prices are residualised on the hedonic controls and neighbourhood and period effects, the change in each city's mean residual between 2024 and October 2026 is computed, and that change is regressed on a Caspian indicator, with p-values from 5,000 random reassignments of the Caspian label across cities (Young, 2019).

4.2 Struck neighbourhoods and their neighbours

The within-Tehran design follows the stacked difference-in-differences approach (Cengiz *et al.*, 2019; Wing *et al.*, 2024). For each struck neighbourhood s, a stack is formed from s and the up to four nearest neighbourhoods that were struck in neither war and lie within 2.5 km. The 49 stacks contain 116 distinct control neighbourhoods, at a median distance of 1.3 km. The regression is

$$\ln p_{instk} = \delta \left(\text{Struck}_n \times \text{Post}_t\right) + X_i'\gamma + \alpha_{sn} + \lambda_{st} + \varepsilon_{instk}, \tag{2}$$

with stack-by-neighbourhood effects α_{sn} and stack-by-period effects λ_{st}, and standard errors clustered by neighbourhood, since one control can appear in several stacks. The stack-by-period effects confine each comparison to neighbourhoods a short distance apart and absorb any shock common to that part of the city, including the wars' citywide effects. Because treatment occurs at a single date for all stacks, the estimator avoids the problems of two-way fixed-effects regressions with staggered treatment (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).

The design is preferred to a comparison of struck neighbourhoods with the rest of Tehran because the struck neighbourhoods are richer and more northerly than the city as a whole, and the price gap between north and south Tehran widened over the period; a citywide comparison would attribute that divergence to the strikes. Robustness checks vary the number of neighbours and the distance cap, add price-tier-by-period effects, restrict treatment to the first war, replace the binary treatment with distance to the nearest struck neighbourhood, and add a struck-specific linear trend. A placebo test that assigns a false post-period of September–December 2024 asks whether the design finds effects where none should exist. Because pre-trend tests have limited power (Roth, 2022), the trend-adjusted estimates are reported alongside the main ones rather than as an afterthought (Rambachan and Roth, 2023).

The same structure, with listing counts per neighbourhood and month as the outcome and Poisson pseudo-maximum likelihood (Santos Silva and Tenreyro, 2006; Correia *et al.*, 2020), describes changes in listing volumes. All models are estimated with the pyfixest package (The PyFixest Authors, 2026).

5 Results

5.1 Rental-market adjustments in northern cities

Table II presents the estimated changes in rental markets across cities. Between 2024 and October 2026, log asking rents per square metre in the Caspian cities increased by 0.241 (s.e. 0.035) relative to Tehran and Alborz, equivalent to approximately 27 per cent. The wild-bootstrap p-value is 0.076. The interactions estimated within 2024 do not indicate statistically significant differences in pre-conflict trends ($p = 0.115$). The estimated rental premium is also positive when the comparison group is restricted to Tehran (0.194, bootstrap $p = 0.008$) or to cities outside the capital region with no recorded direct exposure (0.268, bootstrap $p = 0.004$). The latter estimate corresponds to an increase of approximately 31 per cent. Comparing the Caspian cities with Alborz and Tehran's satellite towns alone produces a larger coefficient of 0.336, but the pre-trend test rejects equality of trends ($p =< 0.001$), making this comparison less suitable for interpreting the post-conflict difference. Rents in Tehran and Alborz changed relatively little compared with the inland cities (0.027), suggesting that the estimated premium is primarily associated with differences in the northern housing markets.

The city-level estimates in Panel C provide a complementary assessment that does not depend on the substantially larger number of listings in Tehran. When cities receive equal weight, the estimated Caspian rental premium relative to the capital region is 0.341 (permutation $p =< 0.001$). Weighting cities by their number of listings produces a coefficient of 0.233 ($p = 0.002$). Relative to inland cities, the corresponding estimates are 0.260 and 0.259, with permutation p-values of 0.010 and 0.018, respectively. Figure 2 illustrates the geographic pattern underlying these estimates. Hedonic-adjusted rents increased by between 17 and 28 per cent across the four Caspian cities, and all 11 cities in the capital region experienced smaller changes than the Caspian city with the lowest increase. The consistency across cities and weighting schemes suggests that the rental differential is not driven solely by the relative size of particular listing markets, although inference remains limited by the small number of northern cities.

An important consideration is whether the Caspian cities were already experiencing faster rental growth before the conflicts. To examine this possibility, a Caspian-specific linear trend is estimated using the 2024 observations and extrapolated to October 2026. This adjustment reduces the estimated rental premium to 0.135 (s.e. 0.049) relative to the capital region and 0.161 (s.e. 0.079) relative to inland cities, corresponding to approximately 14 and 17 per cent. The 2024 coefficients suggest a modest upward differential in Caspian rents, as shown in Figure 3. However, extending a trend estimated over eight months for an additional 22 months requires a strong assumption about how relative rental markets would otherwise have evolved. The trend-adjusted estimates are therefore interpreted as a sensitivity check rather than a definitive correction. The results indicate that pre-existing differences in rental growth may explain part of the observed premium, but they do not eliminate the positive differential.

A separate concern is that three of the four Caspian cities experienced one recorded incident of direct military activity. Restricting the northern sample to Gorgan, the only city without recorded direct exposure, produces an estimated coefficient of 0.239, similar in magnitude to the baseline result. Rental changes in the other three cities are also broadly comparable. This

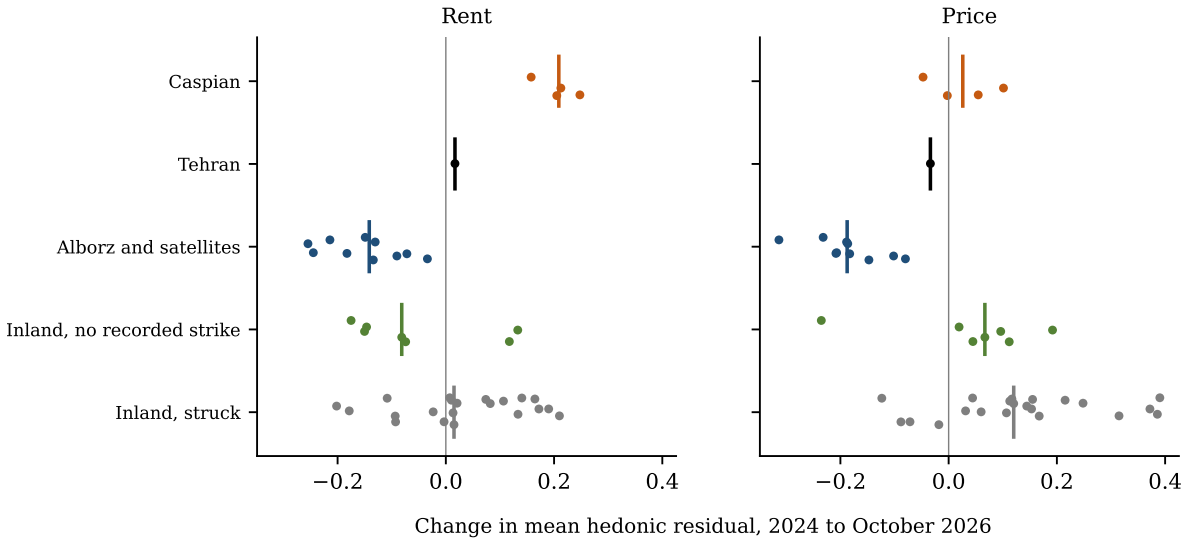


Figure 2: Changes in hedonic-adjusted rents and sale prices by city, 2024 to October 2026
Notes: Each point represents a city with at least 30 listings in both periods. Vertical bars indicate group medians. Changes are calculated from differences in city-level mean residuals obtained from regressions of log asking prices or rent-equivalents per m^2 on observed property characteristics and neighbourhood and period fixed effects.

pattern suggests that the observed premium is not confined to northern cities with recorded incidents. However, with only one city in the restricted sample, statistical inference is limited, and the comparison cannot separately identify the effects of relocation-related demand and other local conditions.

The available archived listings provide additional evidence on the timing of the rental differential. Figure 3 reports the estimated Caspian premium across the periods for which listings are available. The differential is relatively small during 2024 and becomes positive in the archived observations from July 2025. The July estimate is 0.089 (s.e. 0.030), based on 57 listings from the Caspian cities and 91 from the capital region. Restricting the post-conflict observations to the five weeks following the restoration of internet access in June 2025 yields a comparable estimate of 0.099 (s.e. 0.040, bootstrap $p = 0.084$; Supplementary Table A1). The premium subsequently eased before becoming more pronounced in the October 2026 observations. This timing is consistent with an increase in rental demand associated with population movements following the first conflict period. Nevertheless, the archived samples are small and unevenly distributed over time, so they provide suggestive rather than conclusive evidence about the timing and persistence of the underlying demand changes.

Sale prices exhibit a different pattern. Relative to Tehran and Alborz, Caspian sale asking prices increased by 0.116 (bootstrap $p = 0.106$). However, the estimated change relative to inland cities is 0.003 (s.e. 0.067), and the corresponding city-level estimate is also close to zero. Sale prices in Tehran and Alborz were relatively weaker than those in inland cities (−0.104), although this difference is not statistically significant. Thus, the evidence for a Caspian sale-price premium depends on the comparison group, unlike the more consistent positive rental differential. Relative to inland markets, the approximately 30 per cent increase in Caspian rents without a comparable change in sale prices corresponds to a reduction of roughly 23 per cent in the relative asking-price-to-rent ratio. This divergence is consistent with a demand increase

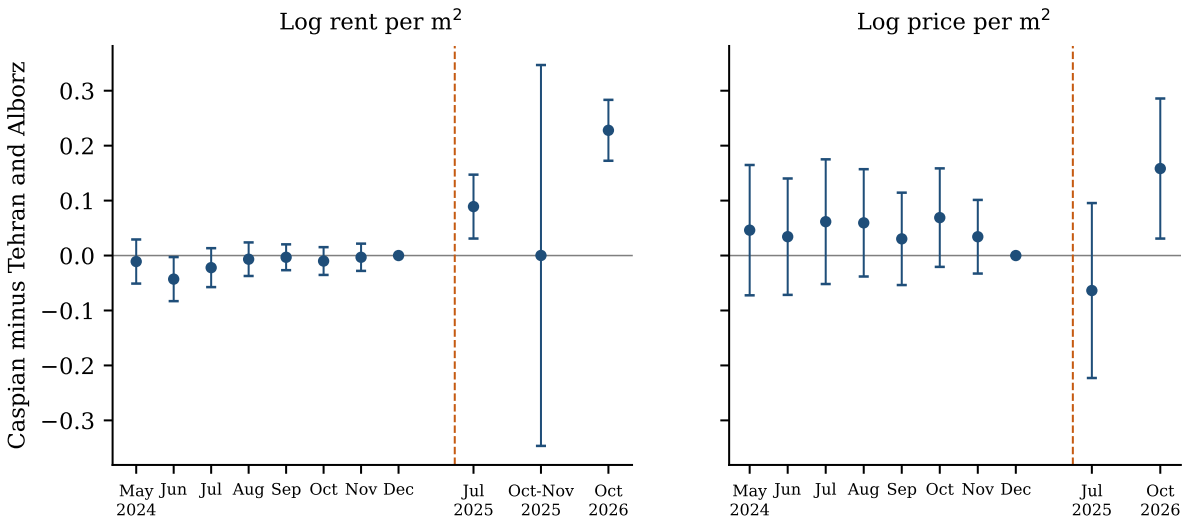


Figure 3: Estimated Caspian rental and sale-price differentials relative to Tehran and Alborz
Notes: Points represent coefficients on Caspian × period from equation (1), with December 2024 as the omitted reference period. Vertical bars indicate 95 per cent confidence intervals based on standard errors clustered by city. July 2025 and October–November 2025 use archived listings. Periods with fewer than 20 archived listings in either group are omitted. Consequently, no archived sale-price estimates are available after July 2025. The dashed line separates the pre-conflict observations.

perceived as temporary, although differences in the adjustment of rental and sale listings provide an alternative explanation.

The magnitude of the rental differential is also economically relevant. The median monthly rent-equivalent in the Caspian sample during 2024 was 160 thousand toman per square metre. Applying the estimated premium of 27 per cent to this baseline implies an additional monthly rent-equivalent of approximately 3.3 million toman for a 75 m^2 apartment. Using the trend-adjusted estimate gives approximately 1.7 million toman. These calculations illustrate the magnitude of the relative rental premium using 2024 asking rents as a common benchmark; they should not be interpreted as changes in the actual rent paid by an individual household.

5.2 Housing-market adjustments within Tehran

Table III presents the estimated changes in asking prices and rents within Tehran. In the preferred specification, asking sale prices per square metre in neighbourhoods with recorded direct exposure changed by 0.011 relative to nearby neighbourhoods without recorded exposure (s.e. 0.013). The corresponding 95 per cent confidence interval ranges from −1.3 to 3.6 per cent. The estimated rental change is 0.022 (s.e. 0.010). Neither estimate provides statistically significant evidence of a local difference in advertised housing prices. Tests using the 2024 observations also do not reject equality of pre-conflict trends ($p = 0.154$ for sale prices and $p = 0.325$ for rents), although the available pre-period is relatively short.

The baseline estimates are similar under several alternative specifications. Adding price-tier-by-period fixed effects produces coefficients of 0.013 for sale prices and 0.012 for rents. The estimates also change little when the matching procedure uses two or six neighbouring comparison areas instead of four, or when the exposed group is restricted to neighbourhoods identified in the June 2025 records. These results suggest that the absence of a clear local

difference in the baseline specification is not driven by a particular choice of nearby comparison neighbourhoods. However, statistical insignificance does not establish that direct exposure had no effect on local housing values.

Two additional analyses qualify the interpretation. First, a placebo comparison between the first and second halves of 2024 indicates that asking rents in neighbourhoods subsequently classified as exposed increased by 0.011 relative to their nearby comparison areas (bootstrap $p = 0.036$). This difference is economically meaningful relative to the small post-conflict rental estimate and suggests that some of the observed change may reflect pre-existing differences in neighbourhood trends. Consequently, the estimated post-conflict rental differential should not be interpreted independently of these earlier movements.

Second, neighbourhoods with recorded exposure are allowed to follow a separate linear trend estimated from the 2024 data. The estimated differential trends are 0.27 per cent per month for sale prices and 0.31 per cent per month for rents. Extrapolating these trends to October 2026 produces coefficients of -0.056 (s.e. 0.043) for sale prices and -0.057 (s.e. 0.032) for rents. The corresponding lower confidence bounds are approximately -13 and -11 per cent. These adjustments yield more negative point estimates than the baseline specification, although they remain imprecisely estimated. Their interpretation depends on whether the differential trends observed during 2024 would have continued over the subsequent 22 months. The event-study estimates in Supplementary Figure A1 do not establish a clear pre-conflict trend, but the limited observation window cannot rule out such a pattern. Taken together, the results provide no clear evidence of a substantial local discount in asking prices, while leaving open the possibility of moderate reductions.

The geographic extent of any local rental response is also difficult to identify. Supplementary Table A1 compares neighbourhoods at different distances from recorded exposure locations. With price-tier-by-period effects included, asking rents in neighbourhoods within 3 km of an exposed location declined by approximately 6 to 8 per cent relative to more distant areas. The directly exposed neighbourhoods show changes of a similar magnitude. However, removing the price-tier interactions reverses the estimated pattern. This sensitivity suggests that differences in housing-market trends across Tehran's price segments play an important role in the distance-based comparisons. The evidence therefore does not make it possible to distinguish reliably between a geographically broader response around exposed neighbourhoods and pre-existing differences across parts of the city.

5.3 Proximity to security-related sites and listing activity

The study also examines whether housing-market changes differed near security-related facilities identified in the geographic data. If proximity to these locations affected expectations about future neighbourhood conditions, it could be reflected in asking prices or rents even in areas without recorded direct exposure. Panel B of Table IV presents these estimates. Within Tehran, after controlling for recorded direct exposure, asking sale prices near these facilities changed by -0.9 per cent and rents by -1.2 per cent, with lower confidence bounds of -4.4 and -3.3 per cent, respectively. Across all 43 cities, the corresponding estimates are -0.3 per cent for sale prices and -1.0 per cent for rents. Neither using continuous distance to the nearest facility

nor constructing stacked comparisons between nearby neighbourhoods with and without such facilities produces a substantially different result. There is therefore no clear evidence of a relative discount associated with this measure of proximity, although the estimates do not rule out smaller or more localized adjustments.

Panel A examines changes in the number of observed property listings. Relative to nearby comparison neighbourhoods, areas with recorded direct exposure had 17 per cent fewer sale listings and 20 per cent fewer rental listings in October 2026 compared with the 2024 sample. However, the placebo analysis indicates that their relative listing counts had already declined between the first and second halves of 2024, by 8 per cent for sale listings and 10 per cent for rental listings.

These quantity estimates require particular caution because the two principal datasets capture different aspects of listing activity. The 2024 observations represent newly posted advertisements over several months, whereas the October 2026 observations represent the available inventory at specific collection dates. Differences in listing duration, renewal frequency and the composition of advertisers can therefore affect the measured changes independently of actual housing-market activity. Moreover, the earlier decline observed in the 2024 placebo suggests that some of the differences may predate the conflict periods. The results are consistent with changes in the relative availability of advertised properties, but they do not establish whether homeowners withdrew properties from the market or whether the underlying supply of housing changed. established by these data.

6 Discussion

6.1 Interpreting the housing-market responses

The findings point to different housing-market adjustments across locations and market segments. Within Tehran, there is no clear difference in asking prices between neighbourhoods directly exposed to military activity and their nearby comparison areas. Nor is there a detectable discount associated with proximity to security-related sites identified in the 2026 records. In contrast, the Caspian cities experienced a substantial increase in asking rents relative to the comparison markets. This difference was already apparent in the limited observations following the first conflict period in 2025. Together, these findings suggest that housing-market adjustments were more pronounced in potential relocation destinations than in neighbourhoods with recorded direct exposure, at least along the margins observed in the data.

One possible explanation is that the two settings experienced different changes in housing demand. Within Tehran, geographically concentrated exposure may not have substantially changed the relative attractiveness of nearby neighbourhoods, particularly if buyers and tenants did not interpret individual incidents as changes in broader neighbourhood conditions. In the Caspian cities, household relocation may have increased demand for rental accommodation at a time when the available housing stock could not adjust quickly. These mechanisms are consistent with the estimates, although the absence of direct measures of relocation and housing-market expectations prevents them from being identified conclusively.

The difference between rental and sale markets provides further insight. Caspian asking

rents increased substantially, while sale asking prices showed no comparable increase relative to inland cities. In a standard asset-pricing framework, this pattern is consistent with an increase in demand that buyers expect to be temporary: current rents respond to additional demand for housing services, whereas property values depend on expected future returns (Poterba, 1984; Himmelberg *et al.*, 2005). The initial rental premium in 2025, followed by a partial easing, is also consistent with this interpretation. However, slower adjustment of sale prices, changes in housing supply, or differences in the required return on residential property could produce similar patterns. Moreover, Caspian sale prices increased relative to Tehran and Alborz, indicating that the observed divergence also depends on the comparison market. The evidence therefore supports a distinction between rental and ownership responses but does not establish the expected duration of additional housing demand.

These findings are consistent with broader evidence that housing-market responses depend on the nature and location of economic changes. Previous research documents property-price adjustments in areas exposed to persistent security risks (Besley and Mueller, 2012; Elster *et al.*, 2017), while studies of refugee and population inflows find increases in rents in receiving markets, with effects that vary across locations and housing segments (Tumen, 2016; Depetris-Chauvin and Santos, 2018; Gluszak and Trojanek, 2025). This paper brings these two dimensions together. They suggest that the geographic distribution of housing-market adjustments need not coincide with the locations of direct exposure, and that rental and sale prices may respond differently to changes in housing demand.

6.2 Limitations

Several limitations are important for interpreting these findings. First, the analysis relies on advertised sale prices and rents rather than completed transactions or rents actually paid. Listing data provide timely information about housing-market conditions and may anticipate changes in transaction-price indices (Anenberg and Laufer, 2017; Lyons, 2019). However, asking prices are not necessarily the prices at which properties are ultimately sold or rented. Sellers choose and revise their listing prices strategically (Knight, 2002; Han and Strange, 2016), while differences in the composition of listed and transacted properties can affect hedonic price estimates (Kolbe *et al.*, 2021). Loss aversion and financial constraints may further influence sellers' willingness to adjust prices or complete transactions (Genesove and Mayer, 2001; Anenberg, 2011). If sale asking prices adjust slowly to weaker demand, the estimated discount in directly exposed Tehran neighbourhoods may understate changes in transaction prices. Differences in the speed of adjustment between rental and sale listings could also contribute to the observed divergence in Caspian markets. Moreover, asking rents for newly advertised properties do not necessarily reflect changes in rents paid by existing tenants.

Second, the 2024 and October 2026 datasets differ in how listings were collected. The earlier observations represent a flow of newly posted advertisements, whereas the October 2026 observations are snapshots of available listings, with a substantial share posted by estate agencies. Hedonic controls and fixed effects account for observed property characteristics and common differences across periods, but they cannot fully address changes in unobserved listing composition that differ between locations. For example, changes in the share of furnished properties, short-

term accommodation, or higher-quality units could affect measured rental differences. Similarly, differences in the frequency with which advertisements are renewed or updated may influence the comparison between rental and sale markets.

Third, the cross-city analysis includes only four Caspian cities, three of which had one recorded incident of direct exposure. Although the city-level permutation tests account for the limited number of treated locations when assessing statistical significance, they do not eliminate concerns about the geographic representativeness of the sample. The city without recorded direct exposure also exhibits a rental premium, suggesting that the pattern is not confined to locations with recorded incidents. Nevertheless, the small number of cities limits the ability to distinguish relocation-related demand from other local changes. The archived listings used to examine the timing of the 2025 premium are also limited, and the estimated short-run dynamics should be interpreted cautiously. Expanding the sample to additional northern cities, including Sari, Amol, Babol and western Mazandaran, would provide a broader assessment of these patterns.

Finally, the geographic classification of direct exposure relies on publicly available reports. Some neighbourhoods identified in reports may not have experienced direct exposure within the boundaries used in this analysis, while other affected neighbourhoods may have been omitted. This concern is particularly relevant to the 2026 period, when the available reports provided less detailed neighbourhood information. Such measurement error could attenuate estimated local differences under some conditions, although its direction and magnitude are not known. The relative stability of the estimates when the analysis is restricted to the first conflict period, for which location reporting is more detailed, provides some reassurance. Nevertheless, the within-Tehran estimates should not be interpreted as evidence that direct exposure had no effect on local housing values. In particular, the sensitivity to pre-existing trends and the imprecision of the estimates leave room for moderate price reductions.

Taken together, these limitations suggest that the findings of this study are best interpreted as evidence of different changes in advertised housing-market conditions across locations and market segments. The pronounced Caspian rental premium is consistent with additional demand associated with household relocation, while the weaker sale-price response may reflect expectations about the persistence of that demand. However, listing-market dynamics, changes in property composition and other local factors remain plausible explanations. The evidence identifies where the most pronounced relative changes in asking rents and sale prices occurred, while leaving the precise mechanisms for further investigation.

7 Conclusion

This paper examines how housing markets adjusted across locations during Iran's 2025–2026 conflicts. This study finds that the most pronounced changes occurred in rental markets in northern cities that were potential destinations for households relocating from Tehran. Asking rents in these cities increased by approximately 27 per cent relative to Tehran and Alborz and 31 per cent relative to inland comparison cities, although allowing for differential pre-existing trends reduces the estimated premium. In contrast, neighbourhoods with recorded direct exposure in Tehran showed no statistically detectable discount relative to nearby areas, although moderate price reductions cannot be excluded.

An important finding is that rental and sale markets responded differently. While Caspian asking rents increased substantially, sale asking prices showed no comparable increase relative to inland cities. This divergence is consistent with a temporary increase in housing demand, but it may also reflect slower adjustment of sale prices or changes in housing supply and listing composition. The available evidence does not make it possible to distinguish conclusively between these explanations. Nevertheless, the results suggest that changes in housing demand associated with population movements may be reflected more strongly in rental markets than in residential asset values, at least over the periods observed.

These findings have implications for how housing-market conditions are assessed during periods of population relocation. Changes in rental affordability may extend beyond locations directly exposed to military activity, making conditions in potential receiving markets relevant alongside those in directly affected areas. Further research using transaction prices, more extensive geographic coverage and direct measures of household relocation would help establish the persistence and underlying mechanisms of these adjustments. More broadly, the evidence highlights the importance of examining both the geographic distribution of housing demand and the distinction between rental and ownership markets when assessing how housing markets respond to sudden population movements.

References

Abadie, A. and Gardeazabal, J. (2003), "The economic costs of conflict: a case study of the Basque Country", *American Economic Review*, Vol. 93 No. 1, pp. 113–132, doi: 10.1257/000282803321455188.

Abadie, A. and Dermisi, S. (2008), "Is terrorism eroding agglomeration economies in central business districts? Lessons from the office real estate market in downtown Chicago", *Journal of Urban Economics*, Vol. 64 No. 2, pp. 451–463, doi: 10.1016/j.jue.2008.04.002.

Accetturo, A., Manaresi, F., Mocetti, S. and Olivieri, E. (2014), "Don't stand so close to me: the urban impact of immigration", *Regional Science and Urban Economics*, Vol. 45, pp. 45–56, doi: 10.1016/j.regsciurbeco.2014.01.001.

Al Jazeera (2026), "Up to 3.2 Million People Displaced across Iran amid US–Israeli Attacks: UN", available at: https://www.aljazeera.com/news/2026/3/12/up-to-3-2-million-people-displaced-across-iran-amid-us-israeli-attacks-un (accessed 11 October 2026).

Anari, A. and Kolari, J. (2002), "House prices and inflation", *Real Estate Economics*, Vol. 30 No. 1, pp. 67–84, doi: 10.1111/1540-6229.00030.

Anenberg, E. (2011), "Loss aversion, equity constraints and seller behavior in the real estate market", *Regional Science and Urban Economics*, Vol. 41 No. 1, pp. 67–76, doi: 10.1016/j.regsciurbeco.2010.08.003.

Anenberg, E. and Laufer, S. (2017), "A more timely house price index", *Review of Economics and Statistics*, Vol. 99 No. 4, pp. 722–734, doi: 10.1162/REST_a_00634.

Atuesta, L.H. and Carrasco, M. (2024), "How frequent and visible criminal violence affects housing prices: evidence from mexico city (2007–2011)", *International Journal of Housing Markets and Analysis*, Vol. 17 No. 5, pp. 1203–1224, doi: 10.1108/ijhma-02-2023-0020.

Bernstein, A., Gustafson, M.T. and Lewis, R. (2019), "Disaster on the horizon: the price effect of sea level rise", *Journal of Financial Economics*, Vol. 134 No. 2, pp. 253–272, doi: 10.1016/j.jfineco.2019.03.013.

Besley, T. and Mueller, H. (2012), "Estimating the peace dividend: the impact of violence on house prices in Northern Ireland", *American Economic Review*, Vol. 102 No. 2, pp. 810–833, doi: 10.1257/aer.102.2.810.

Brakman, S., Garretsen, H. and Schramm, M. (2004), "The strategic bombing of German cities during World War II and its impact on city growth", *Journal of Economic Geography*, Vol. 4 No. 2, pp. 201–218, doi: 10.1093/jeg/4.2.201.

Callaway, B. and Sant'Anna, P.H.C. (2021), "Difference-in-differences with multiple time periods", *Journal of Econometrics*, Vol. 225 No. 2, pp. 200–230, doi: 10.1016/j.jeconom.2020.12.001.

Cameron, A.C., Gelbach, J.B. and Miller, D.L. (2008), "Bootstrap-based improvements for inference with clustered errors", *Review of Economics and Statistics*, Vol. 90 No. 3, pp. 414–427, doi: 10.1162/rest.90.3.414.

Cengiz, D., Dube, A., Lindner, A. and Zipperer, B. (2019), "The effect of minimum wages on low-wage jobs", *Quarterly Journal of Economics*, Vol. 134 No. 3, pp. 1405–1454, doi: 10.1093/qje/qjz014.

Correia, S., Guimarães, P. and Zylkin, T. (2020), "Fast Poisson estimation with high-dimensional fixed effects", *The Stata Journal*, Vol. 20 No. 1, pp. 95–115, doi: 10.1177/1536867X20909691.

Davis, D.R. and Weinstein, D.E. (2002), "Bones, bombs, and break points: the geography of economic activity", *American Economic Review*, Vol. 92 No. 5, pp. 1269–1289, doi: 10.1257/000282802762024502.

Davis, L.W. (2004), "The effect of health risk on housing values: evidence from a cancer cluster", *American Economic Review*, Vol. 94 No. 5, pp. 1693–1704, doi: 10.1257/0002828043052358.

Depetris-Chauvin, E. and Santos, R.J. (2018), "Unexpected guests: the impact of internal displacement inflows on rental prices in Colombian host cities", *Journal of Development Economics*, Vol. 134, pp. 289–309, doi: 10.1016/j.jdeveco.2018.05.006.

Dericks, G.H. and Koster, H.R.A. (2021), "The billion pound drop: the Blitz and agglomeration economies in London", *Journal of Economic Geography*, Vol. 21 No. 6, pp. 869–897, doi: 10.1093/jeg/lbaa022.

Divar (2025), "Divar Real Estate Ads Dataset", available at: https://huggingface.co/datasets/divarofficial/real_estate_ads (accessed 10 October 2026).

Elster, Y., Zussman, A. and Zussman, N. (2017), "Rockets: the housing market effects of a credible terrorist threat", *Journal of Urban Economics*, Vol. 99, pp. 136–147, doi: 10.1016/j.jue.2017.02.003.

Elster, Y., Zussman, A. and Zussman, N. (2019), "Effective counter-terrorism: rockets, Iron Dome, and the Israeli housing market", *Journal of Policy Analysis and Management*, Vol. 38 No. 2, pp. 308–337, doi: 10.1002/pam.22114.

Euronews (2025), "Tehran Residents Flee as Israel–Iran Conflict Continues for Fifth Day", available at: https://www.euronews.com/2025/06/17/tehran-residents-flee-as-israel-iran-conflict-continues-for-fifth-day (accessed 11 October 2026).

Ferman, B. and Pinto, C. (2019), "Inference in differences-in-differences with few treated groups and heteroskedasticity", *Review of Economics and Statistics*, Vol. 101 No. 3, pp. 452–467, doi: 10.1162/rest_a_00759.

Gautier, P.A., Siegmann, A. and Van Vuuren, A. (2009), "Terrorism and attitudes towards minorities: the effect of the Theo van Gogh murder on house prices in Amsterdam", *Journal of Urban Economics*, Vol. 65 No. 2, pp. 113–126, doi: 10.1016/j.jue.2008.10.004.

Genesove, D. and Mayer, C. (2001), "Loss aversion and seller behavior: evidence from the housing market", *Quarterly Journal of Economics*, Vol. 116 No. 4, pp. 1233–1260, doi: 10.1162/003355301753265561.

Gholipour, H.F., Lee, K.O., Taoum, K. and Farzanegan, M.R. (2026), "Housing market functioning and emergency housing governance under armed conflict: descriptive evidence from Iran, 2025–2026", *Applied Economics*, pp. 1–13, doi: 10.1080/00036846.2026.2743810.

Glaeser, E.L. and Gyourko, J. (2005), "Urban decline and durable housing", *Journal of Political Economy*, Vol. 113 No. 2, pp. 345–375, doi: 10.1086/427465.

Gluszak, M. and Trojanek, R. (2025), "War in Ukraine, the refugee crisis, and the Polish housing market", *Housing Studies*, Vol. 40 No. 5, pp. 1043–1065, doi: 10.1080/02673037.2024.2334822.

Goodman-Bacon, A. (2021), "Difference-in-differences with variation in treatment timing", *Journal of Econometrics*, Vol. 225 No. 2, pp. 254–277, doi: 10.1016/j.jeconom.2021.03.014.

Hallstrom, D.G. and Smith, V.K. (2005), "Market responses to hurricanes", *Journal of Environmental Economics and Management*, Vol. 50 No. 3, pp. 541–561, doi: 10.1016/j.jeem.2005.05.002.

Han, L. and Strange, W.C. (2016), "What is the role of the asking price for a house?", *Journal of Urban Economics*, Vol. 93, pp. 115–130, doi: 10.1016/j.jue.2016.03.008.

Himmelberg, C., Mayer, C. and Sinai, T. (2005), "Assessing high house prices: bubbles, fundamentals and misperceptions", *Journal of Economic Perspectives*, Vol. 19 No. 4, pp. 67–92, doi: 10.1257/089533005775196769.

Human Rights Activists News Agency (2025), "Twelve Days Under Fire: A Comprehensive Report on the Iran–Israel War, and Daily War Reports, 14–25 June 2025", available at: https://www.en-hrana.org (accessed 10 October 2026).

Human Rights Activists News Agency (2026), "Daily Reports on the U.S.–Israeli War on Iran, 28 February–8 April 2026, and Statistical Overviews", available at: https://www.en-hrana.org (accessed 10 October 2026).

Internet Archive (2026), "Wayback Machine Captures of divar.ir Listing Pages, 2024–2026", available at: https://web.archive.org (accessed 17 October 2026).

Iran International (2025), "Iran Restores Communications after Wartime Blackout, Minister Apologizes", 25 June 2025, available at: https://www.iranintl.com/en/202506259885 (accessed 26 October 2026).

Jewish Institute for National Security of America (2025), "U.S. and Israeli Strikes Against Iranian Targets", available at: https://jinsa.org/jinsa_report/israeli-strikes-against-iranian-targets/ (accessed 10 October 2026).

Knight, J.R. (2002), "Listing price, time on market, and ultimate selling price: causes and effects of listing price changes", *Real Estate Economics*, Vol. 30 No. 2, pp. 213–237, doi: 10.1111/1540-6229.00038.

Kocornik-Mina, A., McDermott, T.K.J., Michaels, G. and Rauch, F. (2020), "Flooded cities", *American Economic Journal: Applied Economics*, Vol. 12 No. 2, pp. 35–66, doi: 10.1257/app.20170066.

Kolbe, J., Schulz, R., Wersing, M. and Werwatz, A. (2021), "Real estate listings and their usefulness for hedonic regressions", *Empirical Economics*, Vol. 61 No. 6, pp. 3239–3269, doi: 10.1007/s00181-020-01992-3.

Linden, L. and Rockoff, J.E. (2008), "Estimates of the impact of crime risk on property values from Megan's laws", *American Economic Review*, Vol. 98 No. 3, pp. 1103–1127, doi: 10.1257/aer.98.3.1103.

Lyons, R.C. (2019), "Can list prices accurately capture housing price trends? Insights from extreme markets conditions", *Finance Research Letters*, Vol. 30, pp. 228–232, doi: 10.1016/j.frl.2018.10.004.

MacKinnon, J.G. and Webb, M.D. (2018), "The wild bootstrap for few (Treated) clusters", *The Econometrics Journal*, Vol. 21 No. 2, pp. 114–135, doi: 10.1111/ectj.12107.

Miguel, E. and Roland, G. (2011), "The long-run impact of bombing Vietnam", *Journal of Development Economics*, Vol. 96 No. 1, pp. 1–15, doi: 10.1016/j.jdeveco.2010.07.004.

Narayanasamy, A., Panta, H. and Devkota, M. (2026), "Capitalizing consumer school ratings into residential listing prices: evidence from greater houston", *International Journal of Housing Markets and Analysis*, pp. 1–30, doi: 10.1108/ijhma-07-2026-0273.

Ngo, T., Wu, H., Tsui, K., Squires, G. and Wang, X. (2025), "House prices during the COVID-19 pandemic: the impact of "panic" returnees migrants to new zealand", *International Journal of Housing Markets and Analysis*, Vol. 18 No. 7, pp. 24–45, doi: 10.1108/ijhma-03-2025-0055.

OpenStreetMap contributors (2026), "OpenStreetMap Data Extract for Iranian Cities", available at: https://www.openstreetmap.org (accessed 10 October 2026).

Ortega, F. and Taşpınar, S. (2018), "Rising sea levels and sinking property values: hurricane Sandy and New York's housing market", *Journal of Urban Economics*, Vol. 106, pp. 81–100, doi: 10.1016/j.jue.2018.06.005.

Pinto Hernández, F., Rodríguez Iglesias, I. and Moreno Adalid, A.M. (2026), "The impact of the 2023 housing law on rental prices in catalonia: an empirical analysis using differences-in-differences", *International Journal of Housing Markets and Analysis*, Vol. 19 No. 3, pp. 619–636, doi: 10.1108/ijhma-10-2024-0160.

Poterba, J.M. (1984), "Tax subsidies to owner-occupied housing: an asset-market approach", *The Quarterly Journal of Economics*, Vol. 99 No. 4, pp. 729–752, doi: 10.2307/1883123.

Qin, B., Peng, Y. and Feng, L. (2023), "The impacts of pandemic on urban housing prices: evidence from the outbreak of COVID-19 in beijing, 2020", *International Journal of Housing Markets and Analysis*, Vol. 16 No. 3, pp. 474–489, doi: 10.1108/ijhma-08-2022-0112.

Rambachan, A. and Roth, J. (2023), "A more credible approach to parallel trends", *Review of Economic Studies*, Vol. 90 No. 5, pp. 2555–2591, doi: 10.1093/restud/rdad018.

Roback, J. (1982), "Wages, rents, and the quality of life", *Journal of Political Economy*, Vol. 90 No. 6, pp. 1257–1278, doi: 10.1086/261120.

Roodman, D., Nielsen, M., MacKinnon, J.G. and Webb, M.D. (2019), "Fast and wild: bootstrap inference in Stata using boottest", *The Stata Journal*, Vol. 19 No. 1, pp. 4–60, doi: 10.1177/1536867X19830877.

Rosen, S. (1974), "Hedonic prices and implicit markets: product differentiation in pure competition", *Journal of Political Economy*, Vol. 82 No. 1, pp. 34–55, doi: 10.1086/260169.

Roth, J. (2022), "Pretest with caution: event-study estimates after testing for parallel trends", *American Economic Review: Insights*, Vol. 4 No. 3, pp. 305–322, doi: 10.1257/aeri.20210236.

Sá, F. (2015), "Immigration and house prices in the UK", *The Economic Journal*, Vol. 125 No. 587, pp. 1393–1424, doi: 10.1111/ecoj.12158.

Saiz, A. (2003), "Room in the kitchen for the melting pot: immigration and rental prices", *Review of Economics and Statistics*, Vol. 85 No. 3, pp. 502–521, doi: 10.1162/003465303322369687.

Saiz, A. (2007), "Immigration and housing rents in American cities", *Journal of Urban Economics*, Vol. 61 No. 2, pp. 345–371, doi: 10.1016/j.jue.2006.07.004.

Santos Silva, J.M.C. and Tenreyro, S. (2006), "The log of gravity", *Review of Economics and Statistics*, Vol. 88 No. 4, pp. 641–658, doi: 10.1162/rest.88.4.641.

Sun, L. and Abraham, S. (2021), "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects", *Journal of Econometrics*, Vol. 225 No. 2, pp. 175–199, doi: 10.1016/j.jeconom.2020.09.006.

The National (2026), "Internet Restored to Tens of Millions in Iran after Three-Month Blackout", 27 May 2026, available at: https://www.thenationalnews.com/news/mena/2026/05/27/internet-restored-to-tens-of-millions-in-iran-after-three-month-blackout/ (accessed 26 October 2026).

The PyFixest Authors (2026), "pyfixest: Fast High-Dimensional Fixed Effect Estimation in Python", available at: https://github.com/py-econometrics/pyfixest (accessed 26 October 2026).

Trojanek, R. and Gluszak, M. (2022), "Short-run impact of the Ukrainian refugee crisis on the housing market in Poland", *Finance Research Letters*, Vol. 50, p. 103236, doi: 10.1016/j.frl.2022.103236.

Tumen, S. (2016), "The economic impact of Syrian refugees on host countries: quasi-experimental evidence from Turkey", *American Economic Review*, Vol. 106 No. 5, pp. 456–460, doi: 10.1257/aer.p20161065.

Webb, M.D. (2023), "Reworking wild bootstrap-based inference for clustered errors", *Canadian Journal of Economics*, Vol. 56 No. 3, pp. 839–858, doi: 10.1111/caje.12661.

Wing, C., Freedman, S. and Hollingsworth, A. (2024), "Stacked difference-in-differences", NBER Working Paper No. 32054, National Bureau of Economic Research, doi: 10.3386/w32054.

Young, A. (2019), "Channeling Fisher: randomization tests and the statistical insignificance of seemingly significant experimental results", *Quarterly Journal of Economics*, Vol. 134 No. 2, pp. 557–598, doi: 10.1093/qje/qjy029.

Appendix

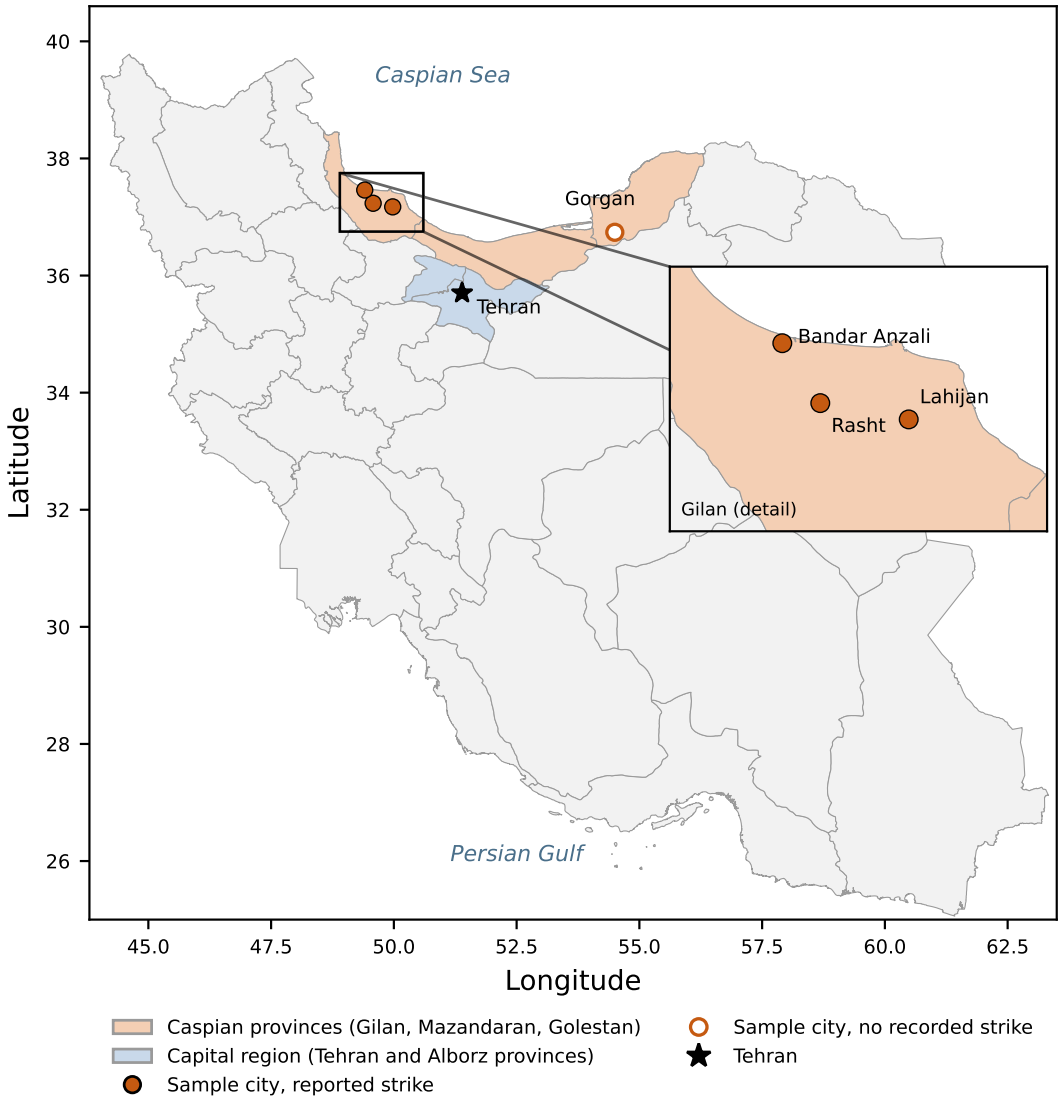


Figure A1: Location of the Caspian sample cities

Notes: Shaded provinces mark the Caspian region (Gilan, Mazandaran and Golestan) and the capital region (Tehran and Alborz). Points mark the four Caspian cities with sufficient listings in the October 2026 sample: Rasht, Bandar Anzali and Lahijan in Gilan Province, and Gorgan in Golestan Province. Filled points denote cities with at least one strike recorded by the sources described in the main text; the hollow point (Gorgan) denotes the city with no recorded strike. The inset enlarges western Gilan, where three of the four cities lie within about 60 km of one another.

Source: Province boundaries from geoBoundaries (gbOpen, IRN ADM1; Runfola *et al.*, 2020), derived from OpenStreetMap data (© OpenStreetMap contributors, ODbL 1.0); city locations and strike records from the authors' data.

Tables

Table I: Listings by group and sample

	Cities	Listings		Median per m^2		Median	Median
		Sale	Rent	Price	Rent	size (m^2)	age (yrs)
Panel A. May–December 2024 (Divar open data)							
Tehran, struck neighbourhoods	1	25,967	24,013	111.8	432	100	9
Tehran, neighbouring controls	1	26,518	19,348	87.0	381	82	10
Tehran, other neighbourhoods	1	28,910	18,866	64.6	288	75	10
Caspian provinces	101	18,012	11,626	31.5	160	96	6
Alborz and Tehran satellite cities	35	65,144	37,658	29.1	151	84	6
Inland cities, no recorded strike	202	23,445	18,937	19.0	90	90	7
Inland cities, struck	43	102,304	74,150	32.8	144	100	7
Panel B. October 2026 (crawl and tracking rounds)							
Tehran, struck neighbourhoods	1	3,862	4,225	360.8	1000	105	10
Tehran, neighbouring controls	1	4,755	4,374	265.6	828	85	11
Tehran, other neighbourhoods	1	4,735	3,623	180.0	606	77	10
Caspian provinces	4	1,852	1,563	108.6	481	100	6
Alborz and Tehran satellite cities	10	6,514	5,503	81.0	300	87	6
Inland cities, no recorded strike	7	3,017	2,602	66.2	217	90	6
Inland cities, struck	21	12,726	10,766	104.0	315	105	7

Notes: Apartments of 25–400 m^2 after trimming (Section 3). Prices in million toman per m^2; rents are monthly rent-equivalents (rent plus three per cent of the deposit) in thousand toman per m^2. Neighbouring controls are the up to four nearest unstruck neighbourhoods within 2.5 km of each struck neighbourhood. Panel A covers every city in the 2024 release; the October 2026 collection covers the 43 cities with at least 800 sale listings in 2024. Toman values are nominal; annual inflation exceeded 30 per cent over the period, so only relative comparisons are informative.

Table II: Displacement: change in Caspian housing costs, 2024 to October 2026

	Log rent per m^2			Log price per m^2		
	Estimate	WCR p	Pre p	Estimate	WCR p	Pre p
Panel A. Listing-level difference-in-differences						
Caspian vs Tehran and Alborz	0.241***	0.076	0.115	0.116***	0.106	0.580
	(0.035)			(0.040)		
Caspian vs Tehran	0.194***	0.008	0.140	0.071***	0.088	0.638
	(0.011)			(0.026)		
Caspian vs inland, no recorded strike	0.268***	0.004	0.230	0.003	0.968	0.331
	(0.048)			(0.067)		
Caspian cities without a recorded strike vs Tehran and Alborz[a]	0.239	n.a.	–	0.191	n.a.	–
	(0.034)			(0.029)		
Caspian vs Alborz and satellites	0.336***	0.002	<0.001	0.188***	0.024	0.478
	(0.035)			(0.039)		
Tehran and Alborz vs inland, no recorded strike	0.027	0.707	0.028	−0.104	0.230	0.342
	(0.056)			(0.069)		
Panel B. Adding a group-specific linear trend fitted on 2024						
Caspian vs Tehran and Alborz	0.135***			0.220		
	(0.049)			(0.209)		
Caspian vs inland, no recorded strike	0.161**			−0.263		
	(0.079)			(0.259)		
Panel C. City-level two-step estimates (permutation p in place of WCR p)						
Caspian vs Tehran and Alborz, equal weights	0.341	<0.001		0.198	0.001	
	(0.032)			(0.039)		
Caspian vs Tehran and Alborz, listing weights	0.233	0.002		0.107	0.202	
	(0.033)			(0.044)		
Caspian vs inland, no recorded strike, equal weights	0.260	0.010		−0.016	0.834	
	(0.053)			(0.061)		
Caspian vs inland, no recorded strike, listing weights	0.259	0.018		0.024	0.918	
	(0.055)			(0.083)		

Notes: Panel A: coefficient on treated group × October 2026 from a regression of log price (or rent-equivalent) per m^2 on hedonic controls, neighbourhood fixed effects and period fixed effects; standard errors clustered by city in parentheses; WCR p is the wild cluster restricted bootstrap p-value (Webb weights, 499 draws); Pre p is the joint test that the treated-by-month interactions for May–November 2024 are zero. Panel B adds the treated group's linear monthly trend, identified from 2024 and extrapolated to October 2026. Panel C: city-level change in mean hedonic residual between 2024 and October 2026 (cities with at least 30 listings in both), regressed on a Caspian indicator; HC1 standard errors; p-values from 5,000 permutations of the Caspian label. *, **, *** denote $p < 0.10$, 0.05, 0.01 for the cluster-robust t-test. The last row of Panel A is a check on the control group, not a displacement estimate. [a]Gorgan only; with a single treated city neither the cluster-robust test nor the bootstrap is informative.

Table III: Struck neighbourhoods versus their nearest unstruck neighbours, Tehran

	Log price per m^2		Log rent per m^2		Struck	Control
	Estimate	WCR p	Estimate	WCR p	nbhds	nbhds
Four nearest within 2.5 km	0.011	0.453	0.022**	0.048	49	116
	(0.013)		(0.010)			
+ price tier × period	0.013	0.343	0.012	0.244	49	116
	(0.011)		(0.009)			
Two nearest within 1.5 km	0.008	–	0.022**	–	48	71
	(0.016)		(0.010)			
Six nearest within 3 km	0.011	–	0.023**	–	50	145
	(0.012)		(0.009)			
Struck in June 2025 only	0.011	0.513	0.021**	0.078	41	100
	(0.014)		(0.010)			
Adding a struck-group linear trend fitted on 2024	−0.056		−0.057*		49	116
	(0.043)		(0.032)			
Placebo: Sep–Dec vs May–Aug 2024	0.010*	0.202	0.011**	0.036	49	116
	(0.006)		(0.005)			
Pre-trend test, first row (p)	0.154		0.325			

Notes: Stacked difference-in-differences. Each stack contains one struck neighbourhood and its nearest unstruck neighbourhoods (centroid distance); neighbourhoods struck in either war are never controls. The coefficient is struck × October 2026, with May–December 2024 as the pre-period, hedonic controls, stack-by-neighbourhood and stack-by-period fixed effects. Standard errors clustered by neighbourhood in parentheses; WCR p as in Table II. The trend row adds struck × month (2024 months 0–7, October 2026 = 29). The placebo row uses 2024 data only. *, **, *** denote $p < 0.10$, 0.05, 0.01.

Table IV: Listing counts and proximity to police and Basij sites

	Sale		Rent	
	Estimate	p	Estimate	p
Panel A. Neighbourhood listing counts, Poisson pseudo-maximum likelihood				
Struck vs nearest neighbours, Oct 2026 vs 2024	−0.192	0.067	−0.223	0.008
	(0.105)		(0.084)	
Placebo: Sep–Dec vs May–Aug 2024	−0.079	0.064	−0.104	0.001
	(0.043)		(0.032)	
Panel B. Police station or Basij base within 500 m, log price or rent per m^2				
Tehran (controls for struck × post)	−0.009	0.627	−0.012	0.269
	(0.018)		(0.011)	
All 43 cities	−0.003	0.860	−0.010	0.335
	(0.016)		(0.010)	
Tehran, exposed vs four nearest unexposed	−0.006	0.659	0.008	0.581
	(0.012)		(0.011)	
Tehran, log(1 + km to nearest site)	0.027	0.667	0.023	0.363
	(0.050)		(0.024)	
All cities, log(1 + km to nearest site)	0.007	0.850	0.021	0.269
	(0.029)		(0.017)	

Notes: Panel A: neighbourhood-by-month listing counts in the stacked sample of Table III, with stack-by-neighbourhood and stack-by-period fixed effects; cluster-robust p. The October 2026 counts come from snapshots of active inventory and the 2024 counts from a flow of new listings, so Panel A is descriptive. Panel B: coefficient on exposure × October 2026 with neighbourhood and city-by-period fixed effects and hedonic controls; standard errors clustered by neighbourhood; p is the WCR bootstrap p-value. Sites are OpenStreetMap features tagged as police stations or Basij bases.